

Mehmet Yusuf Ocak

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Computer Engineering Student | GPA: 3.72 / 4.00 (High Honors Student)

Education

Sivas University of Science and Technology – Sivas, Türkiye **October 2022 – June 2027**
B.Sc. in Computer Engineering (100% English) | Elected Department Representative

Experience

TÜBİTAK 3501 – Sivas, Türkiye **March 2026 – Present**
Researcher in Autonomous Systems and Communications

- Designed a **LoRa**-based RF communication layer and synchronized mission commands, RSSI data, and the ground-station control flow across a multi-UAV swarm using the **MAVLink** protocol.
- Conducted RF signal processing and RSSI-based localization studies in **MATLAB**; integrated swarm behavior, swarm positioning, and autonomous formation missions with **ROS 2** and **ArduPilot** architectures.

TÜBİTAK 3501 – Sivas, Türkiye **July 6, 2026 – August 3, 2026**
Researcher in Autonomous Systems and RF Communications – Mandatory Summer Internship

- Built a multi-UAV communications infrastructure using **Raspberry Pi Zero**, **RTL-SDR**, and **LoRa/SiK** telemetry modules; implemented RSSI collection, time synchronization, and data transfer to the ground station.
- Performed multi-vehicle identification, telemetry configuration, and hardware verification tests in **ArduPilot** and **Mission Planner**.

ArkSigner Yazılım – Ankara, Türkiye **June 2025 – July 2025**
Computer Engineering Intern – Mandatory Summer Internship

- Developed asynchronous **.NET** verification services integrated with **Amazon S3** and low-latency **WebSocket** backend architectures for real-time AI4KYC hologram detection.

Sivas Bilim ve Teknoloji Üniversitesi – Sivas, Türkiye **November 2024 – December 2024**
Docker Product Development Intern

- Implemented containerized architectures with **Docker** and **Docker Hub** to deploy cloud-based distributed systems and synchronize microservices.

İHA Kontrol ve Simülasyon Laboratuvarı – Sivas, Türkiye **October 2024 – November 2024**
Research Intern

- Designed high-fidelity flight simulation test environments using **ROS 2** and **Gazebo** to validate autonomous flight control systems.

Technical Projects

TEKNOFEST Air Defense Systems Competition | ROS 2, OpenCV, PID, C++ **2025 - 2026**

- Turan Teknoloji Team (2025) - 6th Place in Turkey:** Acted as the autonomy team leader in a competition with 515 participating teams. Successfully passed all reporting and live-fire test stages, securing 6th place in the finals.
- AeroGuardians Team (2026) - Finalist:** Successfully passed the Preliminary Design (PDR) and Critical Design (CDR) stages in the TEKNOFEST Celikkubbe Air Defense Systems Competition. Completed the System Capability Video (SCV) evaluation and advanced to the grand finals.
- Autonomous Target Acquisition & Engagement:** Developed the end-to-end software architecture for detecting, continuously tracking, and making autonomous engagement (firing) decisions for moving aerial targets in accordance with competition specifications.
- Visual Tracking (One-Shot Learning):** Instead of relying on heavy, compute-intensive models, integrated a zero-shot tracking algorithm capable of extracting embeddings from a single frame for real-time tracking.
- Orientation & Motion Control (PID):** Generated PID-based orientation commands for mechanical axes utilizing the ROS 2 framework to continuously center and lock onto moving targets.

TEKNOFEST Combat UAV Competition (Team CASA) | ROS, Gazebo, Python, OpenCV **2025**

- Autonomy & Simulation:** Developed interception and kamikaze algorithms within ROS and Gazebo simulation environments to autonomously detect rival UAVs, make tactical dogfight decisions, and lock onto targets.
- Field Testing & Crisis Management:** Successfully completed Hardware-in-the-Loop (HIL) integrations following the Critical Design Report (CDR). Although an unexpected mechanical crash during flight proof tests prevented in-field validation of the autonomy modules, the process provided significant engineering experience regarding hardware limitations and system integration.

Hospital Information System (HIS) Network Design | Cisco, VLAN, OSPF, ASA Firewall **2026**

- Designed a scalable enterprise network architecture by segmenting hospital units into VLANs and subnets for clinical, administrative, server, DMZ, and guest networks.

- Configured **OSPF** for dynamic route exchange in a multi-router topology and **DHCP and DNS** for centralized IP allocation and name resolution services.
- Secured internal, DMZ, and internet traffic with **Cisco ASA, ACL, and NAT** rules; isolated the guest network through a separate router to prevent unauthorized access to the corporate network.

Smart Logistics Route Optimization System | *ML, Optimization, Explainable AI*

2026

- Predicted delay risk at each stop using 39 features with a Stacking Ensemble architecture (LightGBM, XGBoost, CatBoost); optimized using F2-Score and Pinball Loss.
- Implemented dynamic route optimization by integrating ML predictions with a 9-component cost function in Google OR-Tools, achieving **15.6% cost savings**.
- Developed a three-layer hallucination control mechanism with Llama 3.2 and a TF-IDF-based RAG system to provide reliable AI explanations.

ResearchOS - AI-Powered Autonomous Academic Ecosystem | *Django, Next.js, PostgreSQL, MAS, NLP*

2026

- Designed a **Multi-Agent System (MAS)** architecture and **NLP-based** semantic analysis pipeline that autonomously collects and synchronizes data from external sources (Google Scholar, LinkedIn).
- Built an AI-powered decision engine that dynamically matches researcher profiles with strategic funding calls; integrated a **PostgreSQL** and **Django REST** backend with an **SSR-enabled Next.js** frontend.

AI4KYC - Identity Verification System | *Kotlin, .NET, Computer Vision, WebSocket*

2025

- Developed a mobile solution using **Kotlin, C# .NET**, and the **CameraX** API that analyzes holograms on official documents in real time to prevent identity fraud and provides instant user guidance.
- Enabled low-latency, bidirectional data transfer between the image-processing server and mobile application via **WebSocket**; optimized the application architecture using **MVVM** principles.

Amazon S3 Automation Service | *.NET Core, AWS S3, ELK Stack, FFmpeg*

2025

- Built a multithreaded **C# .NET** backend service that automates integrity and playability tests with **FFmpeg** for large video files hosted on **AWS S3** and **MinIO** servers in the banking sector.
- Reduced operational workload through automatic removal of corrupted files; made all verification processes visible and reportable in real time using **Elasticsearch** and **Kibana**.

Autonomous Hardware Traffic Controller | *Digital Logic, FSM, Flip-Flop, PCB Design*

2024

- Designed a deterministic hardware FSM based on Karnaugh Maps and synchronous sequential logic, operating without microcontroller dependency or CPU load.
- Prototyped the system on a breadboard using flip-flops for state transitions and 555 Timer ICs for timing delays.
- Completed the multilayer/double-sided PCB design of the finalized schematic, optimized the copper layout, and prepared Gerber files for manufacturing.

Technical Skills

- **Programming Languages:** C++, C, Python, Java, Kotlin, C# (.NET), SQL
- **Autonomous Systems and Robotics:** ROS 2, ROS/Gazebo, PID Control, Sensor Fusion, OpenCV
- **Artificial Intelligence and Data Science:** YOLO, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas
- **Development Tools and Hardware:** Git, Docker (Docker Hub), Linux, Altium Designer (PCB Design), Amazon S3

Languages

English: C1 - Professional Working Proficiency

Awards and Certificates

- **Yüksek Onur Belgesi** (High Honors Certificate, Fall 2025–2026: 4.00/4.00, SBTÜ)
- **Yüksek Onur Belgesi** (High Honors Certificate, Spring 2025–2026: 3.94/4.00, SBTÜ)
- **TEKNOFEST 2025** – 6th Place in Türkiye (Hava Savunma Sistemleri Kategorisi / Air Defense Systems Competition)
- **Anadolu Hackathon** – 5th Place (2026)
- **Milli Teknoloji Yapay Zekâ Uzmanlık Programı** (2026)
- **SHGM İHA-0 & İHA-1 İnsansız Hava Aracı Pilot Lisansı** (2026)
- **Siber Vatan Siber Güvenlik Eğitim Programı Mezuniyeti** (2025)
- **BAKA Siber Güvenlik CTF Başarı Sertifikası** (2024)
- **Pupilica Android Geliştirme Bootcamp** (2025)

References

Kerem Kemaneci (Bureau Veritas - Cybersecurity Team Lead): keremkemaneci@gmail.com | +31 6 45441703
Yunus Emre Özcan (ASELSAN - Senior Lead Engineer): ye.ozcan@aselsan.com.tr | +90 532 585 73 44